What happens when two protons collide at $\sqrt{s} = 13.6$ TeV?

Minimum-bias pp collisions: Pythia 8 events, a C++/ROOT analysis and a look at ALICE data

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`github.com/tomekstolarczyk/pp-minbias-13p6tev`

Contents

The task

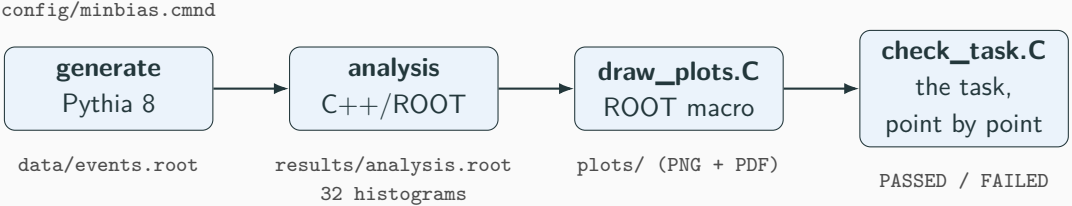
- generate $\sim$100 000 minimum-bias pp collisions at $\sqrt{s} = 13.6$ TeV with Pythia 8
- keep only final-state particles
- histogram the multiplicity N, p_{T}, η and ϕ for all charged particles, $\pi^{\pm}$, $K^{\pm}$, p and $\bar{p}$
- draw every histogram, export PNG and PDF

Steps

1. the setup: generation, analysis, how I read the task
2. results: the 32 histograms
3. comparing the categories with each other
4. comparing the simulation with ALICE data
5. wrapping up: checks and reproducibility

The setup

The pipeline



- `scripts/run_all.sh` runs everything with one command, in about 2 minutes (4 with the ALICE comparison)
- built with CMake as true C++17; Pythia 8.318, ROOT 6.40.04, GCC 12.3, Ubuntu 22.04
- every run leaves its log in `results/`, so every number in this presentation can be traced back

Generation

config/minbias.cmnd	
Beams:idA, idB	2212, 2212 (pp)
Beams:eCM	13600 GeV
SoftQCD:inelastic	on
Tune:pp	14 (Monash 2013)
Main:numberOfEvents	100 000
Random:seed	12345

- only particles with `isFinal()` are stored: PDG code, charge, p_{T}, η, ϕ
- the generator keeps going until exactly 100 000 good events are stored
- 19 265 955 final-state particles in total
- the settings are also stored inside `analysis.root`

Analysis

category	selection
`all`	every final-state particle
`ch`	charge $\neq 0$
`pip`, `pim`	π^+, π^- (PDG ± 211)
`Kp`, `Km`	K^+, K^- (PDG ± 321)
`p`, `pbar`	p, $\bar{p}$ (PDG ± 2212)

histogram	bins	range
`hN_*`	400	$[0, 400]$
`hPt_*`	100	$[0, 10]$ GeV
`hEta_*`	100	$[-5, 5]$
`hPhi_*`	64	$[0, \pi]$

4 quantities $\times$ 8 categories $=$ 32 histograms in `results/analysis.root`. The multiplicity is filled once per event, also when it is zero; p_{T}, η and ϕ once per particle.

How I read the task

Minimum bias `SoftQCD:inelastic`: every inelastic collision, without elastic scattering, which makes no new particles. ALICE's own Pythia setup uses the same switch.

Final state literally: what `isFinal()` says, with Pythia's default decays.

ϕ range Pythia gives $(-\pi, \pi]$, the task asks for $[0, \pi]$: I fill $|\phi|$, which has the same distribution because no direction around the beam is special.

`all` and `ch` `all` = every final-state particle, neutral ones included; `ch` = charge $\neq 0$.

Nothing is thrown away no cuts; values outside a range go to the under- or overflow bin and count in the mean and standard deviation.

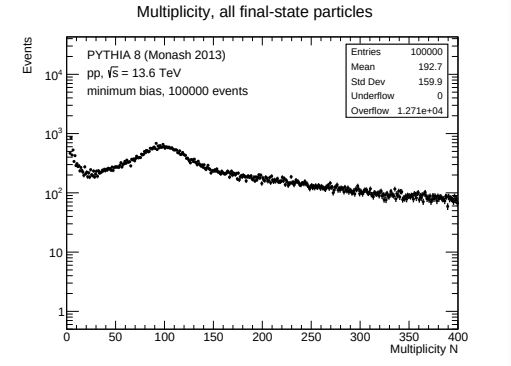
Results: the 32 histograms

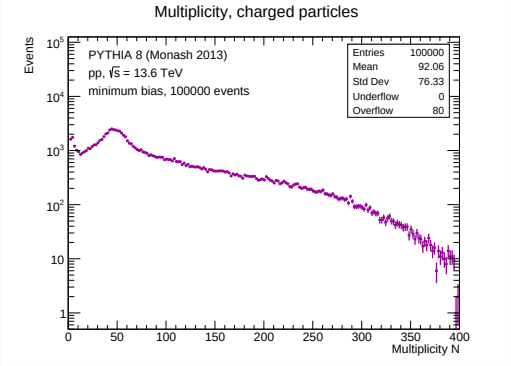
Results at a glance

category	$\langle N \rangle$ per event	$\langle p_{\mathrm{T}} \rangle$ [GeV]	outside $\lvert\eta\rvert < 5$
all	192.66	0.40	25.7%
charged	92.06	0.51	26.8%
π^+ / π^-	38.36 / 37.71	0.48 / 0.48	26.1 / 25.0%
K^+ / K^-	4.24 / 4.12	0.68 / 0.68	27.7 / 26.0%
p / $\bar{p}$	3.82 / 2.60	0.69 / 0.78	50.3 / 29.0%

- pions are 83% of the charged particles, kaons 9%, protons and antiprotons 7%
- the means include the values outside the histogram ranges `results/analysis.log`

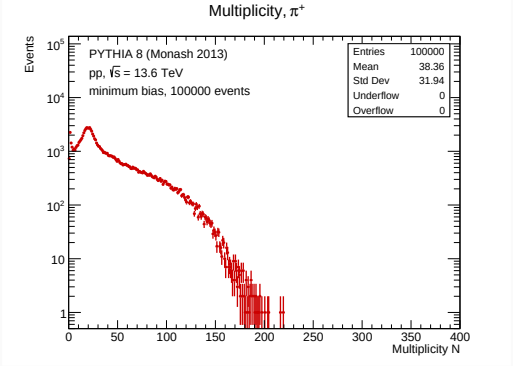
Multiplicity: all and charged particles

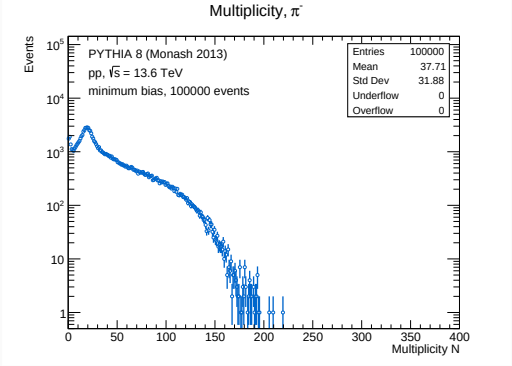




- 192.7 particles per event, 92.1 of them charged; 12.7% of the events have $N_{\mathrm{all}} \geq 400$ and sit in the overflow
- N_{ch} is always even (the two protons have a total charge of $+2$), so every odd bin of `hN_ch` is empty

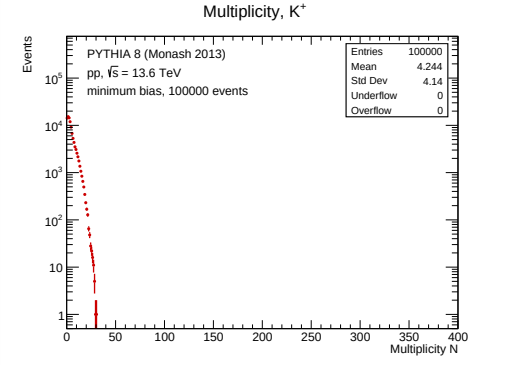
Multiplicity: π^+ and π^-

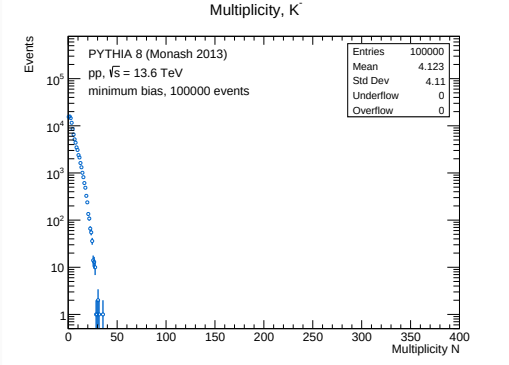




- 38.4 π^+ and 37.7 π^- per event, with a long tail to more than 150
- 731 events have no π^+, 1 755 have no π^-

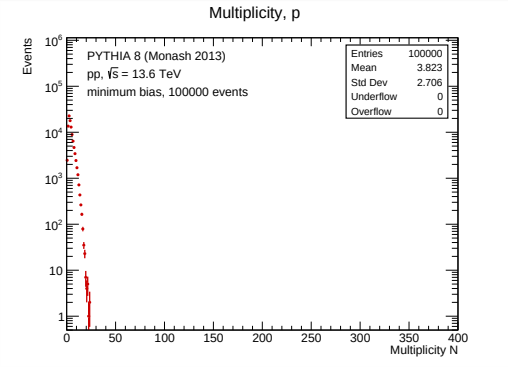
Multiplicity: K^+ and K^-

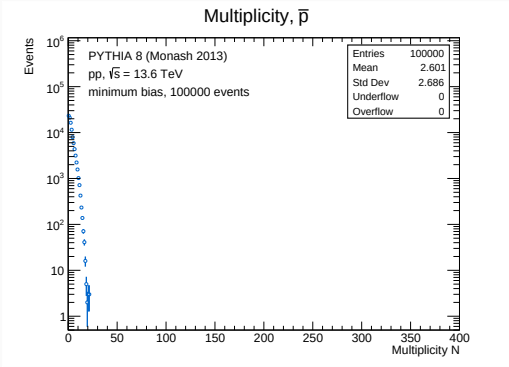




- 4.24 K^+ and 4.12 K^- per event, about 9 times fewer than pions
- 14 254 events have no K^+, 15 474 have no K^-

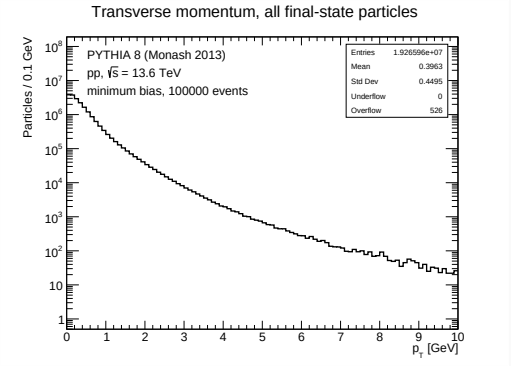
Multiplicity: p and $\bar{p}$

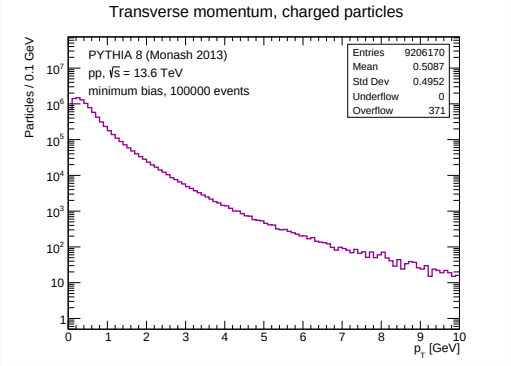




- 3.82 protons but only 2.60 antiprotons per event
- only 2 457 events have no proton, against 23 205 without an antiproton

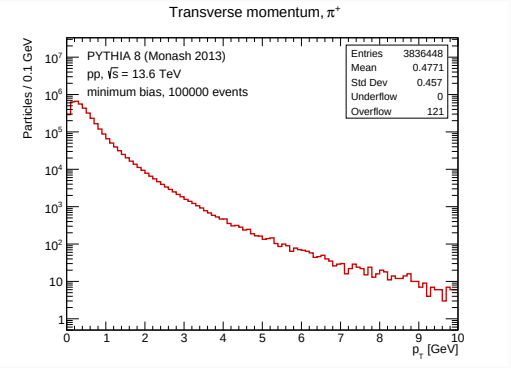
Transverse momentum: all and charged particles

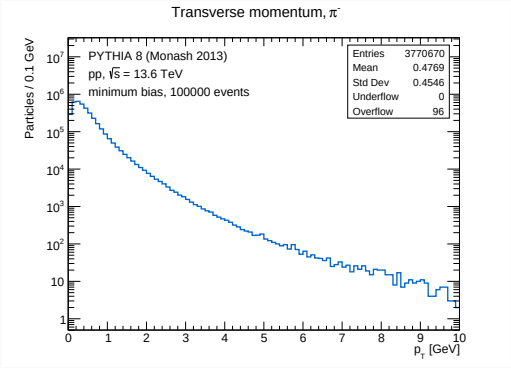




- most particles have p_{T} around 0.25 GeV; up to 10 GeV the spectrum falls by about five orders of magnitude
- $\langle p_{\mathrm{T}} \rangle = 0.40$ GeV for all particles, 0.51 GeV for the charged ones: the neutral ones are softer

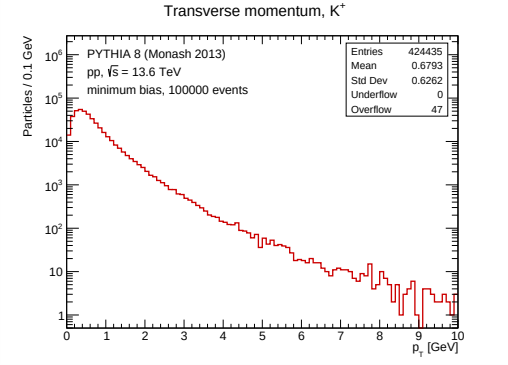
Transverse momentum: π^+ and π^-

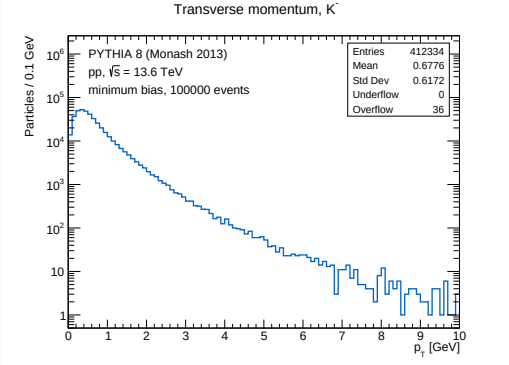




- the two spectra look the same, $\langle p_{\mathrm{T}} \rangle = 0.48$ GeV for both
- 121 π^+ and 96 π^- have $p_{\mathrm{T}} > 10$ GeV and are kept in the overflow

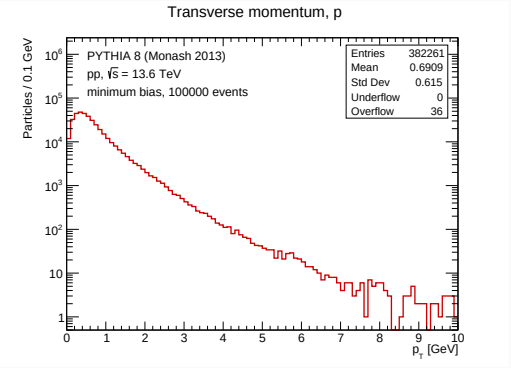
Transverse momentum: K^+ and K^-

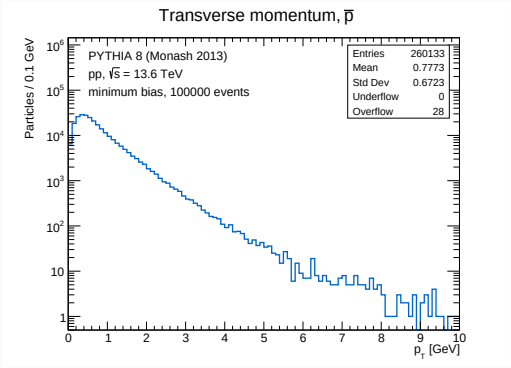




- $\langle p_{\mathrm{T}} \rangle = 0.68$ GeV, higher than for pions; the peak is at 0.35 GeV instead of 0.25 GeV
- with fewer particles, the tail above about 5 GeV fluctuates more

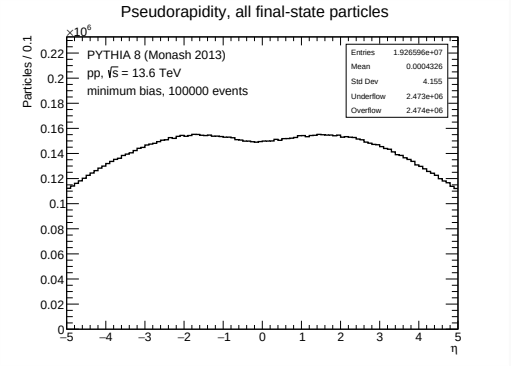
Transverse momentum: p and $\bar{p}$

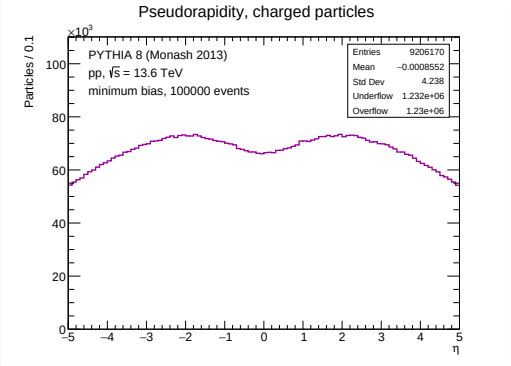




- $\langle p_{\mathrm{T}} \rangle = 0.69$ GeV for protons, 0.78 GeV for antiprotons
- the overlay later shows where the two spectra differ: below about 2 GeV

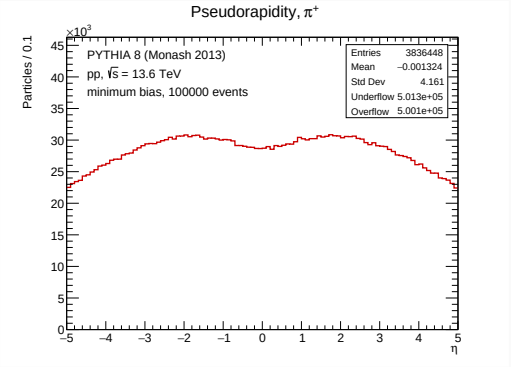
Pseudorapidity: all and charged particles

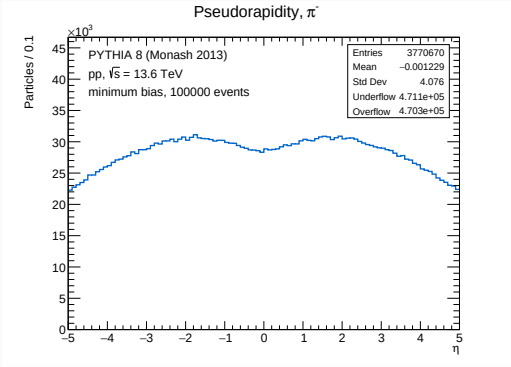




- symmetric around $\eta = 0$, with a shallow dip in the middle
- the task range cuts off a lot: 25.7% of all particles and 26.8% of the charged ones have $|\eta| > 5$

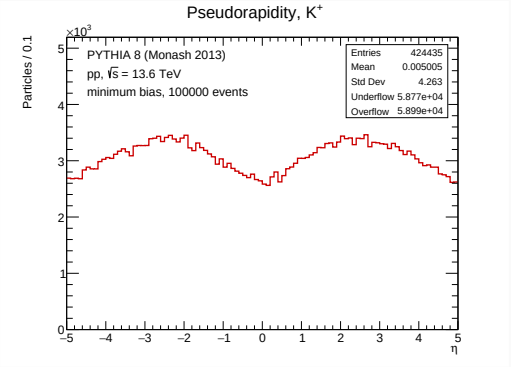
Pseudorapidity: π^+ and π^-

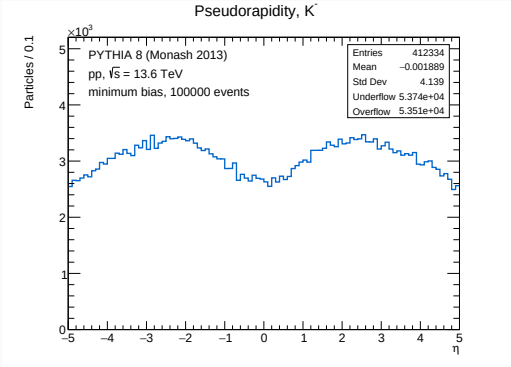




- a broad plateau with a shallow dip at $\eta = 0$
- 26.1% of the π^+ and 25.0% of the π^- are outside $|\eta| < 5$

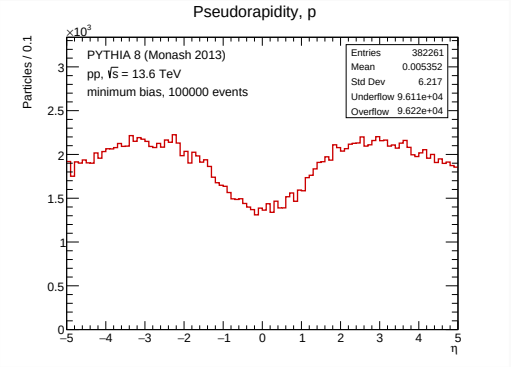
Pseudorapidity: K^+ and K^-

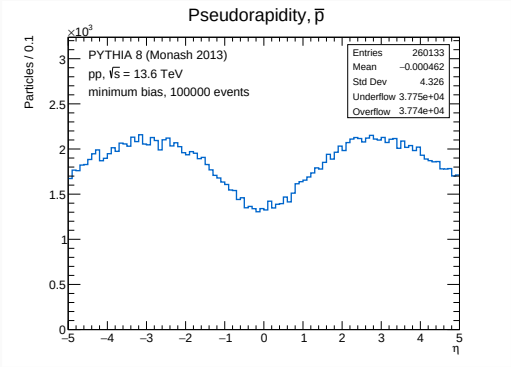




- the dip at $\eta = 0$ is deeper than for pions
- 27.7% of the K^+ and 26.0% of the K^- are outside $|\eta| < 5$

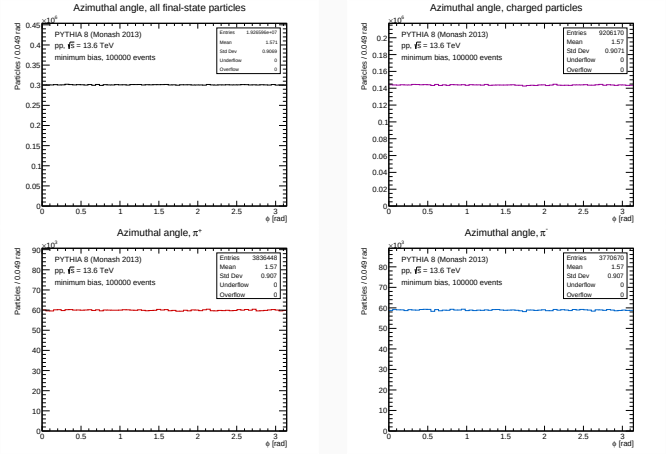
Pseudorapidity: p and $\bar{p}$





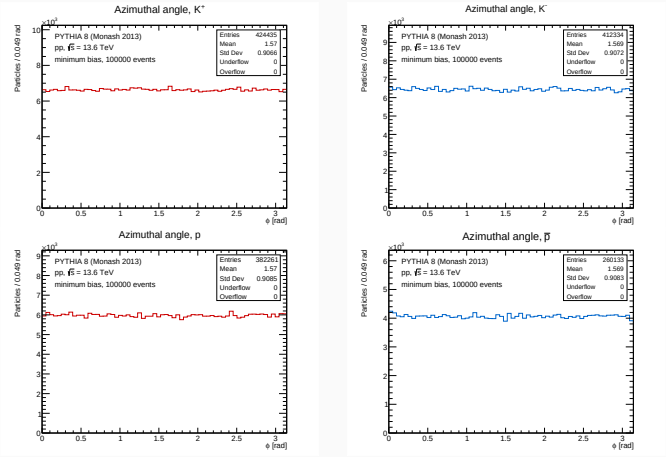
- the deepest dip of the three particle types
- 50.3% of the protons but only 29.0% of the antiprotons are outside $|\eta| < 5$

Azimuthal angle: all, charged, π^+ and π^-



Flat: mean $1.571 = \pi/2$ and standard deviation $0.907 = \pi/\sqrt{12}$, the values of a uniform distribution on $[0, \pi]$. Thanks to $|\phi|$, no particle falls outside the range.

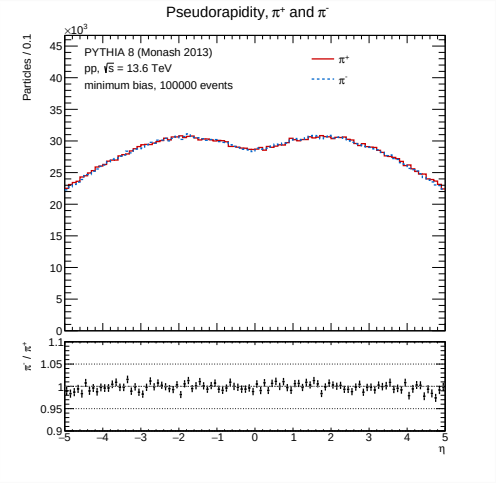
Azimuthal angle: $K^{\pm}$, p and $\bar{p}$



Also flat; with fewer particles per bin, the bin-to-bin fluctuations are larger.

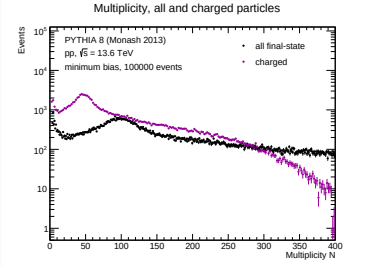
Comparing the categories

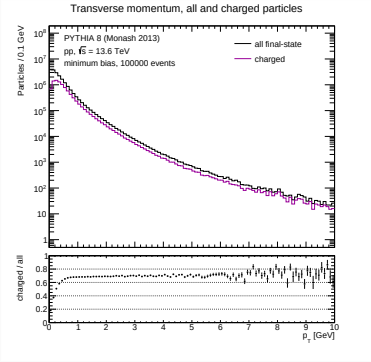
How to read the overlays



- two categories on one canvas: positive solid, negative dashed
- the lower panel shows their ratio, so differences of a few percent become visible
- the error bars come from the bin counts; for charged/all they are binomial, because one category is part of the other
- where the ratio panel is noisy, there are too few particles to say anything, e.g. above about 4 GeV for protons

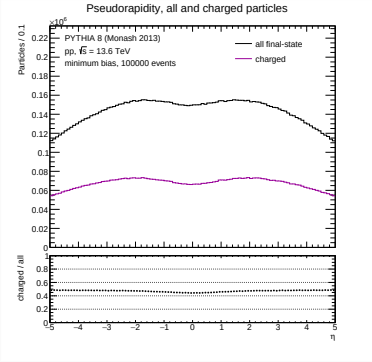
All and charged particles: N and p_{T}

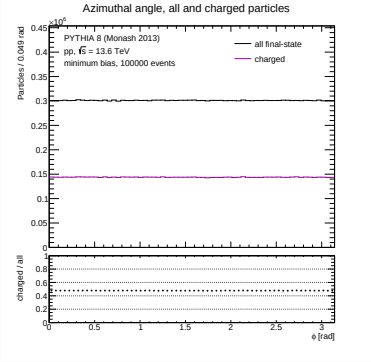




- at the lowest p_{T} only 18% of the particles are charged; from about 1 GeV on, 69%
- in the event tree, 99% of the neutral particles below 0.1 GeV are photons

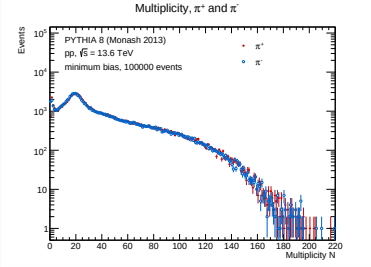
All and charged particles: η and ϕ

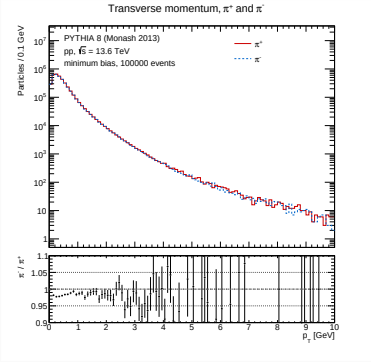




- about 47% of the particles are charged, nearly the same at every η (slightly fewer at $\eta = 0$) and every ϕ

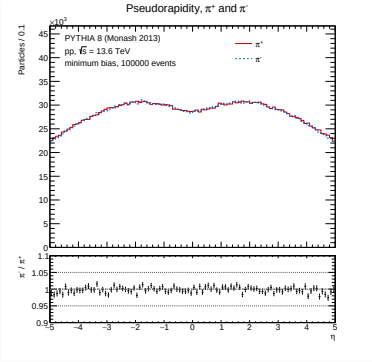
π^+ and π^-: N and p_{T}

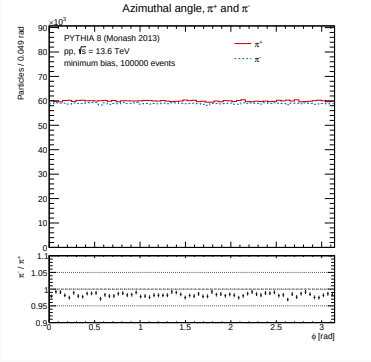




- the multiplicities almost coincide; at $N = 0$ the π^- point is higher
- π^-/π^+ is close to 1 at every p_{T}

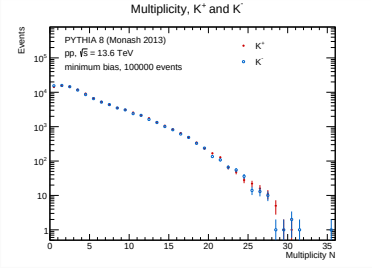
π^+ and π^-: η and ϕ

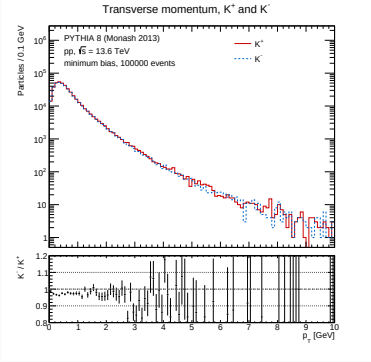




- $\pi^-/\pi^+ \approx 0.98$, flat in η and in ϕ: a small excess of π^+ everywhere

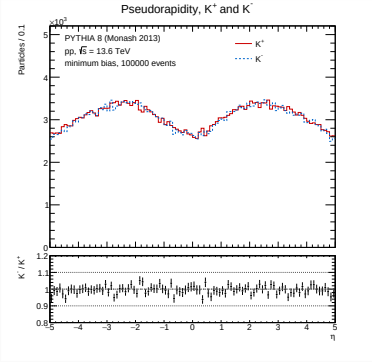
K^+ and K^-: N and p_{T}

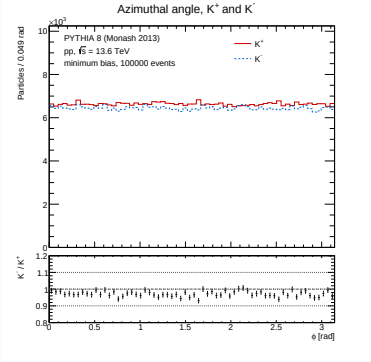




- $K^-/K^+ \approx 0.97$ at low p_{T}
- above a few GeV the ratio is dominated by statistical fluctuations

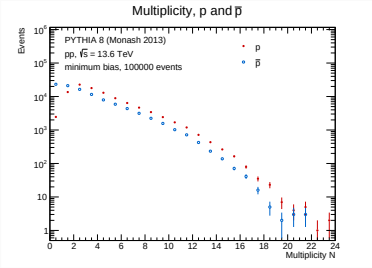
K^+ and K^-: η and ϕ

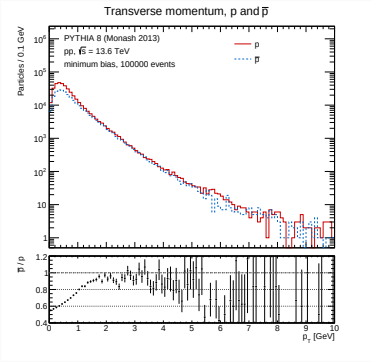




- flat ratios, slightly below 1, in η and in ϕ

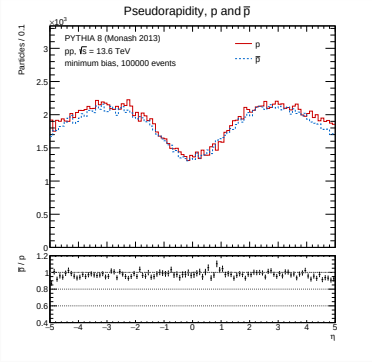
p and $\bar{p}$: N and p_{T}

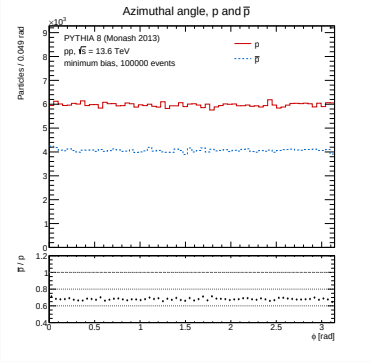




- $\bar{p}/p$ rises from 0.55 at the lowest p_{T} to about 0.95 at 2 GeV: the extra protons are at low p_{T}
- above about 4 GeV there are too few protons for the ratio to mean anything

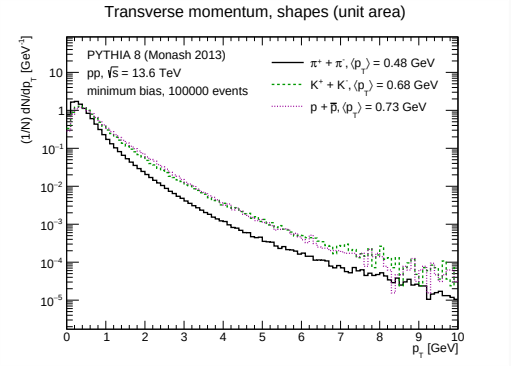
p and $\bar{p}$: η and ϕ

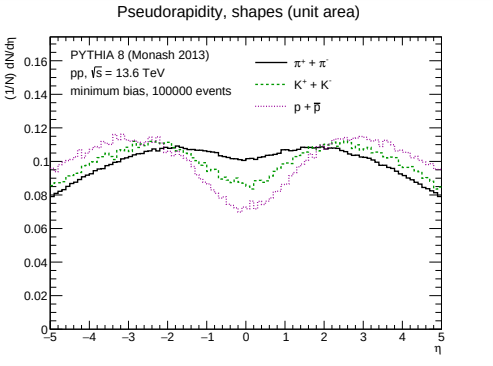




- inside $|\eta| < 5$, $\bar{p}/p = 0.97$ and flat
- so most of the extra protons are outside the range: 50% of the protons against 29% of the antiprotons have $|\eta| > 5$

Shapes: π, K and p normalised to unit area





- each curve divided by its number of particles, so only the shapes are compared
- the mean p_{T} grows with the mass: 0.48, 0.68 and 0.73 GeV for π, K and $p+\bar{p}$
- the dip at $\eta = 0$ gets deeper with the mass: 93%, 77% and 63% of the maximum

Two checks hidden in the histograms

Charge. Two protons have a total charge of $+2$, so 100 000 events must have 200 000 more positive than negative particles. From the histogram entries:

$\pi^+ - \pi^-$	65 778
$K^+ - K^-$	12 101
$p - \bar{p}$	122 128
sum	200 007

The other charged particles (e, μ, ...) make up the remaining -7.

Protons. The colliding particles are protons, and it shows:

- 3.82 p vs 2.60 $\bar{p}$ per event
- 2 457 events without a proton vs 23 205 without an antiproton

A sign mistake in a PDG code would break the charge sum at once, so this is also a check of the category selection.

Comparing with ALICE data

Finding a paper to compare with

A simulation is a model, and a model is only as good as its agreement with real data.

arXiv, ALICE papers with 13.6 TeV in the title:

- D^{*+}-meson spin alignment (2609.39713)
- B^0-meson production cross section (2603.18904)
- time resolution of the TOF detector (2511.10311)
- Inclusive and multiplicity-dependent pseudorapidity densities of charged particles (2602.10658)

Only the last one is about the same particles: the η distribution of charged particles, per event and per unit of η ($\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$). That is my `hEta_ch`, divided by the number of events and the bin width.

Counting the way ALICE counts

Events: the paper reports INEL>0 collisions, with at least one charged particle in $|\eta| < 1$. I apply the same selection.

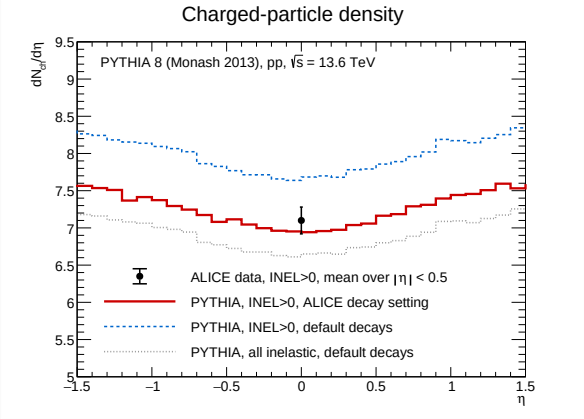
Decays: ALICE's own Pythia configuration (AliceO2, `pythia8_inel.cfg`) lets only particles with $c\tau_0 < 10$ mm decay, so K^0_S, Λ and similar particles stay undecayed. My comparison run does the same (`config/minbias_alice.cmnd`).

Effect of the decay setting, per event:

	default	ALICE
charged	92.06	83.77
π^+	38.36	34.45
K^+	4.24	4.24
p	3.82	3.05

π^+ -10%, p -20%, kaons unchanged.

One definition at a time



$\langle \mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta \rangle$ in $\lvert\eta\rvert < 0.5$:	
task run	6.675
+ only INEL>0	7.713
+ ALICE decay setting	7.004
ALICE measurement	7.10 ± 0.18

With the same definitions, PYTHIA agrees with the measurement within its uncertainty (ratio 0.987).

Multiplicity classes

class (%)	data	PYTHIA (ALICE)	PYTHIA (mine)	mine/ data
0–1	21.78 ± 0.38	25.50	25.96	1.19
1–5	18.48 ± 0.25	20.71	20.91	1.13
5–10	15.76 ± 0.22	17.19	17.44	1.11
10–15	13.89 ± 0.19	14.71	14.66	1.06
15–20	12.50 ± 0.17	12.62	12.58	1.01
20–30	10.86 ± 0.15	10.02	10.10	0.93
30–40	9.09 ± 0.13	7.51	7.60	0.84
40–50	7.63 ± 0.11	5.60	5.64	0.74
50–70	5.87 ± 0.09	3.72	3.76	0.64
70–100	3.69 ± 0.06	2.47	2.49	0.68
0–100	7.10 ± 0.18	7.11	7.00	0.99

$\langle \mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta \rangle$ in $|\eta| < 0.5$, INEL>0; classes as in Table 2 of the paper

ALICE sorts the collisions by the signal in its forward detector FT0 (Table 2 of the paper). I count charged particles in the same η ranges and split my events the same way.

- my PYTHIA matches theirs within 2% in every class
- the average agrees, the extremes do not: 19% too many particles in the busiest class, up to 36% too few in the quiet ones, as the paper also reports

Wrapping up

The task, checked point by point

`tests/check_task.C` runs at the end of every pipeline run:

task	what is checked
technical	build with `-std=c++17`, ROOT 6, Pythia 8
1) generation	pp, `Beams:eCM = 13600`, SoftQCD, 100 000 events, total charge $+2$ in every event
2), 3) histograms	32 names, binning and ranges, one `hN` entry per event, every ϕ in $[0, \pi]$, each category holds exactly its particles
4) macro	opens `analysis.root`, axis labels, legends, every plot as PNG and PDF

Each check was tested by breaking its line of code on purpose (wrong PDG sign, no `isFinal()`, no $|\phi|$, $N = 0$ not filled, wrong number of bins) and watching it fail.

Reproducible by design

- one command, `scripts/run_all.sh`: build, generate, analyse, draw, check, compare
- the random seed is fixed in the `.cmnd` file, and the Pythia settings are stored inside `analysis.root` in a form Pythia can read back
- running again, also in a fresh clone of the repository, gives bin-for-bin identical histograms
- every log is in the repository: Pythia's settings and error statistics, every histogram's mean, standard deviation and out-of-range counts, the ALICE comparison

Summary

- N_{ch} is always even, and there are more π^+, K^+ and p than π^-, K^- and $\bar{p}$
- the extra protons are at low p_{T} and mostly outside $|\eta| < 5$
- heavier particles have a higher mean p_{T} and a deeper dip at $\eta = 0$
- with ALICE's definitions, PYTHIA matches the measured average density at 13.6 TeV, but not the busiest and the quietest collisions
- every step runs with one command, is checked against the task and gives the same histograms every time

Thank you!

github.com/tomekstolarczyk/pp-minbias-13p6tev

ALICE has not published identified-particle spectra at 13.6 TeV yet.
I hope that changes soon, and I would be glad to help :)